Chapter 8

Chemical Composition

Jellybeans

- Suppose you work in a candy store that sells gourmet jelly beans by the bean.
- Can we count these nonidentical beans by weighing?
  - Yes. The key piece of information we need is the average mass of the jelly beans.
- For purposes of counting, the objects behave as though they were all identical, as though they each actually had the average mass.
- The same concept applies in chemistry.

Think, Pair, Share Activity – Unit 8.1

- Can you think of any other items that are counted by weighing? Turn to your neighbor and discuss some examples.

Atoms

- Individual atoms are far too small to see.
- We must learn to count atoms by weighing samples containing large numbers of them.
- Because atoms are so tiny, the normal units of mass—the gram and the kilogram—are much too large to be convenient.

Atomic Mass Units

- Scientists have defined a much smaller unit of mass called the atomic mass unit (amu).

$$1 \text{ amu} = 1.66 \times 10^{-24} \text{ g}$$

Table 8.1 Average Atomic Mass Values for Some Common Elements

Element	Average Atomic Mass (amu)
Hydrogen	1.008
Carbon	12.01
Nitrogen	14.01
Oxygen	16.00
Sodium	22.99
Aluminum	26.98

Elements

- Let’s assume we have a sample of aluminum that has a mass of 26.98 g.
- What mass of copper contains exactly the same number of atoms as this sample of aluminum?
- Because the average masses of aluminum and copper atoms are 26.98 amu and 63.55 amu, respectively, 26.98 g of aluminum and 63.55 g of copper contain exactly the same number of atoms.
- Remember, samples in which the ratio of the masses is the same as the ratio of the masses of the individual atoms always contain the same number of atoms.

The Mole

- The mole (mol) can be defined as the number equal to the number of carbon atoms in 12.01 grams of carbon.
- Techniques for counting atoms very precisely have been used to determine this number to be 6.022×10^{23}.
- This number is called Avogadro's number.
- One mole of something consists of 6.022×10^{23} units of that substance.

Critical Thinking Activity – Unit 8.3

- What if you were offered $1 million to count from 1 to 6×10^{23} at a rate of one number each second?
- Determine your hourly wage. Would you do it? Could you do it?

One Mole

Table 8.2 Comparison of 1-Mole Samples of Various Elements

Element	Number of Atoms Present	Mass of Sample (g)
Aluminum	6.022×10^{23}	26.98
Gold	6.022×10^{23}	196.97
Iron	6.022×10^{23}	55.85
Sulfur	6.022×10^{23}	32.07
Boron	6.022×10^{23}	10.81
Xenon	6.022×10^{23}	131.3



Mole Conversion Example

- How many H atoms are present in a sample with a mass of 0.500 g?

1 mol H atoms = 1.008 g H

$$0.500\ \cancel{\text{g H}} \times \frac{1\ \text{mol H}}{1.008\ \cancel{\text{g H}}} = 0.496\ \text{mol H in sample B}$$

1 mol = 6.022×10^{23}

$$0.496\ \cancel{\text{mol H atoms}} \times \frac{6.022 \times 10^{23}\ \text{H atoms}}{1\ \cancel{\text{mol H atoms}}} = 2.99 \times 10^{23}\ \text{H atoms in sample B}$$

Mole Conversion Practice

- Convert each of the following amounts to moles:

3.60 g Al

T

- 544 g Ca

Atom Conversion Practice

- Determine the number of atoms in each of the following amounts:

0.133 mol Al

Continued practice

- 13.6 mol Ca

Convert atoms to grams

- 2.31×10^{22} atoms of gold

Complex Problems

- Being able to solve complex problems is a talent that will serve you well in all walks of life.
- It is our purpose in this text to help you learn to solve problems in a flexible, creative way based on understanding the fundamental ideas of chemistry.
- We call this approach conceptual problem solving.

Conceptual Problem Solving – Principle 1

- To help us as we proceed to solve a problem, the following organizing principles will be useful to us.

1. First, we need to read the problem and decide on the final goal.
   - Then we sort through the facts given, focusing on keywords and often drawing a diagram of the problem.
   - In this part of the analysis, we need to state the problem as simply and as visually as possible.
   - We can summarize this process as “Where Are We Going?”

Conceptual Problem Solving – Principle 2

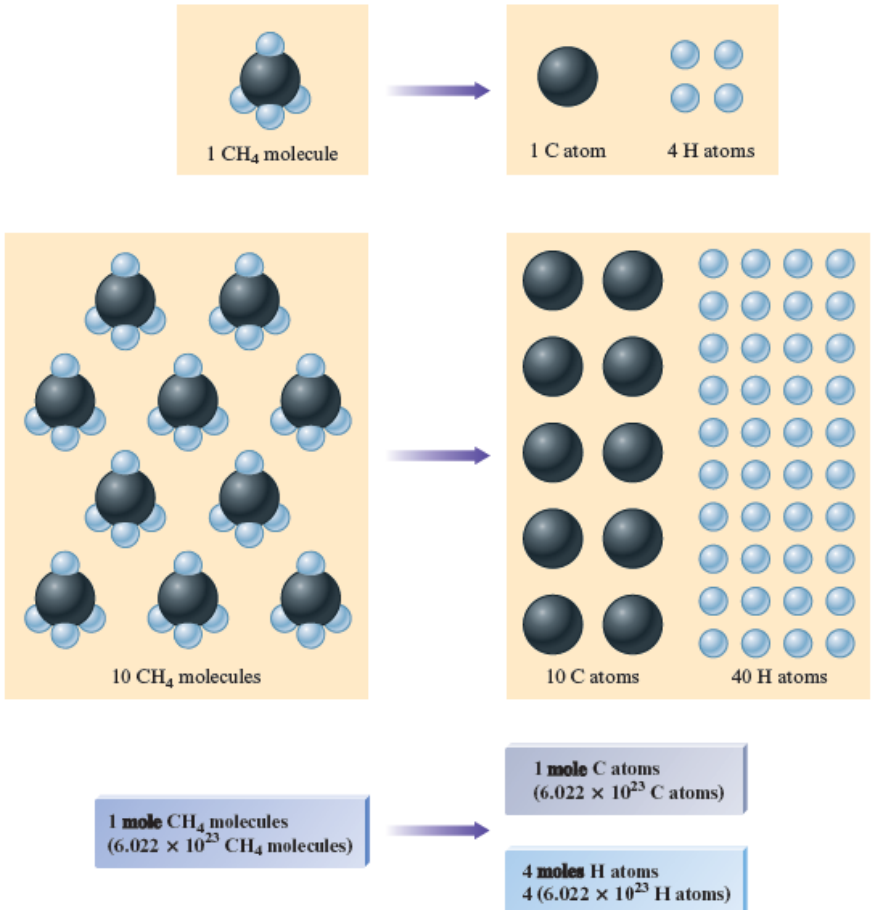
2. We need to work backward from the final goal in order to decide where to start.
   - For example, in a stoichiometry problem we always start with the chemical reaction.
   - Then as we proceed, we ask a series of questions, such as “What are the reactants and products?,” “What is the balanced equation?,” and “What are the amounts of the reactants?”
   - Our understanding of the fundamental principles of chemistry will enable us to answer each of these simple questions and eventually will lead us to the final solution.
   - We can summarize this process as “How Do We Get There?”

Conceptual Problem Solving – Principle 3

3. Once we get the solution of the problem, then we ask ourselves: “Does it make sense?” That is, does our answer seem reasonable?
   - We call this the Reality Check.
   - It always pays to check your answer.

TopHat

From Molecules to Moles



Molar Mass

- The mass of 1 mole of methane can be found by summing the masses of carbon and hydrogen present:

Mass of 1 mol C = 1 × 12.01 g	= 12.01 g
Mass of 4 mol H = 4 × 1.008 g	= 4.032 g
Mass of 1 mol CH_4	= 16.04 g

- The quantity 16.04 g is called the molar mass for methane: the mass of 1 mole of CH_4 molecules.
- The molar mass of any substance is the mass (in grams) of 1 mole of the substance.
  - The molar mass is obtained by summing the masses of the component atoms.

Calculating Molar Mass Practice

- Calculate the molar mass of the following compounds:

NH_3

More practice

- H_2SO_4

Continue to practice

- $C_6H_{12}O_6$

LA problem

- Find the molar mass of barium hydroxide

- $Ba(OH)_2$

- 137.33 + 2(17.01) = 171.35 g/mol

Calculating Mass from Moles Practice

- A sample of NaOH contains 7.89 moles. What is the mass in grams of this sample?

Calculating Moles from Mass Practice

- A sample of Fe_2O_3 weighs 982 g. How many moles does the sample contain?

LA problem 1

- 9.43 mg of calcium chloride.

Mass Percent

- The mass fraction for each element is calculated as follows:

$$\text{Mass fraction for a given element} = \frac{\text{mass of the element present in 1 mole of compound}}{\text{mass of 1 mole of compound}}$$

- The mass fraction is converted to mass percent by multiplying by 100%.

Mass Percent of Ethanol

- Ethanol has the formula C_2H_5OH

$$\text{Mass percent of C} = \frac{\text{mass of C in 1 mol } C_2H_5OH}{\text{mass of 1 mol } C_2H_5OH} \times 100\%$$

$$= \frac{24.02\ \text{g}}{46.07\ \text{g}} \times 100\% = 52.14\%$$

$$\text{Mass percent of H} = \frac{\text{mass of H in 1 mol } C_2H_5OH}{\text{mass of 1 mol } C_2H_5OH} \times 100\%$$

$$= \frac{6.048\ \text{g}}{46.07\ \text{g}} \times 100\% = 13.13\%$$

$$\text{Mass percent of O} = \frac{\text{mass of O in 1 mol } C_2H_5OH}{\text{mass of 1 mol } C_2H_5OH} \times 100\%$$

$$= \frac{16.00\ \text{g}}{46.07\ \text{g}} \times 100\% = 34.73\%$$

Calculating Mass Percent Practice

- What is the mass percent of each element in H_3PO_4?

TopHat 1

- Mass percent for magnesium iodite

Formulas of Compounds

- The formula of a compound represents the relative numbers of the various types of atoms present.
  - For example, the molecular formula CO_2 tells us that for each carbon atom there are two oxygen atoms in each molecule of carbon dioxide.
- Sometimes a chemical reaction gives a product that has never been obtained before.
  - In such a case, a chemist determines what compound has been formed by determining which elements are present and how much of each. These data can be used to obtain the formula of the compound.

What if you are given a percent composition?

- What if a compound has 32.000% carbon, 6.714% hydrogen, 42.630% oxygen, and 18.660% nitrogen?

  Step 1:

  Consider the percentage to be the number of grams in a 100 gram sample:

  32.000 g C

  6.714 g H

  42.630 g O

  18.660 g N

Combustion analysis continued

Step 2:

Find the moles of each element:

$$\frac{32.000\ \text{g}\ \cancel{C}}{12.01\ \text{g}\ \cancel{C}/\text{mol C}} = 2.664\ \text{mol C}$$

$$\frac{6.714\ \text{g}\ \cancel{H}}{1.01\ \text{g}\ \cancel{H}/\text{mol H}} = 6.648\ \text{mol H}$$

$$\frac{42.630\ \text{g}\ \cancel{O}}{16.00\ \text{g}\ \cancel{O}/\text{mol O}} = 2.664\ \text{mol O}$$

$$\frac{18.660\ \text{g}\ \cancel{N}}{14.00\ \text{g}\ \cancel{N}/\text{mol N}} = 1.333\ \text{mol N}$$

Step 3:

Divide each by the smallest number of moles:

$$\frac{2.664 \text{ mol C}}{1.333 \text{ mol}} = 2 \text{ C}$$

$$\frac{6.648 \text{ mol H}}{1.333 \text{ mol}} = 5 \text{ H}$$

$$\frac{2.664 \text{ mol O}}{1.333 \text{ mol}} = 2 \text{ O}$$

$$\frac{1.333 \text{ mol N}}{1.333 \text{ mol}} = 1 \text{ N}$$

$$C_2H_5O_2N$$

Percent composition to formula

- A compound has the following percent composition
- C = 9.94%
- Cl = 58.6%
- F = 31.4%

Empirical and Molecular Formulas

- The formula of a compound that expresses the smallest whole-number ratio of the atoms present is called the empirical formula or simplest formula.
  - A compound that contains the molecules $C_4H_8O_4$ has the same empirical formula as a compound that contains $C_6H_{12}O_6$ molecules.
  - The empirical formula for both is CH_2O.
- The actual formula of a compound—the one that gives the composition of the molecules that are present—is called the molecular formula.

Empirical Formulas Practice

- What are the empirical formulas of the following compounds?

H_2O_2

$C_4H_8O_2$

P_4O_{10}

Calculation of Empirical Formulas

- One of the most important things we can learn about a new compound is its chemical formula.
- To calculate the empirical formula of a compound, we first determine the relative masses of the various elements that are present.
- One way to do this is to measure the masses of elements that react to form the compound.

Steps for Determining the Empirical Formula of a Compound – Steps 1-3

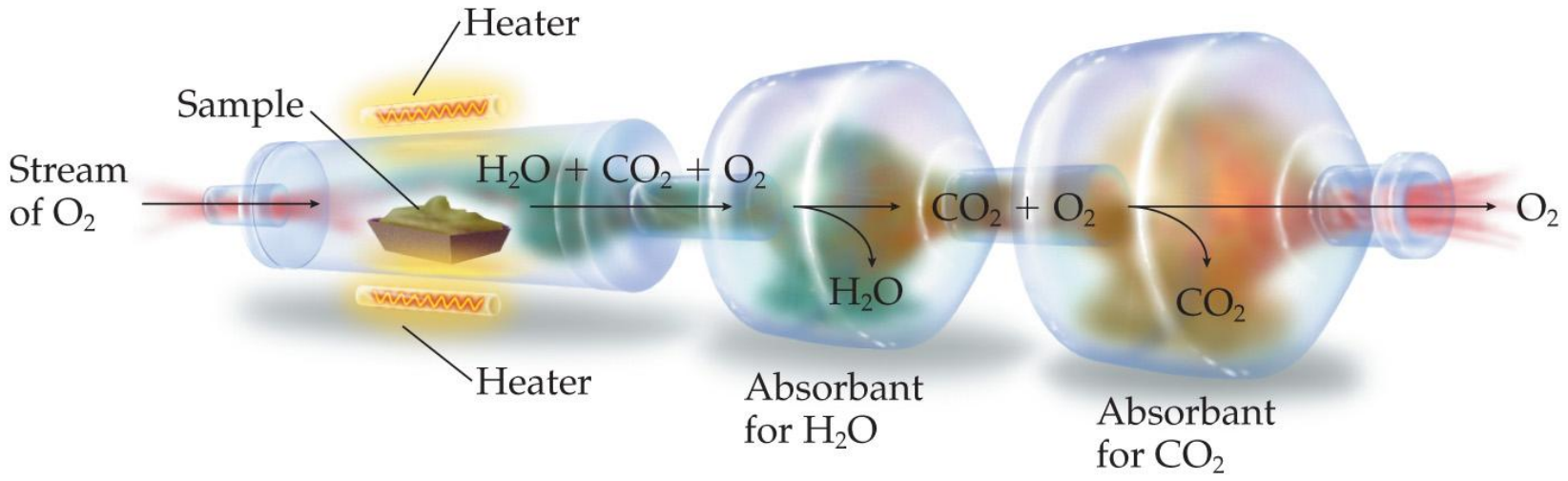
1. Obtain the mass of each element present (in grams).
2. Determine the number of moles of each type of atom present.
3. Divide the number of moles of each element by the smallest number of moles to convert the smallest number to 1.
   - If all of the numbers so obtained are integers (whole numbers), these are the subscripts in the empirical formula.
   - If one or more of these numbers are not integers, go on to step 4.

Steps for Determining the Empirical Formula of a Compound – Step 4

4. Multiply the numbers you derived in step 3 by the smallest integer that will convert all of them to whole numbers.
   - This set of whole numbers represents the subscripts in the empirical formula.

Combustion analysis

- A common technique called combustion analysis is good for determining the empirical formulas of carbon containing compounds. This works by burning the sample in excess oxygen so that it full converts to products.
- When we do this, we can find the percent of carbon and the other elements which leads us to the empirical formula of the material being identified.



Empirical formulas

- Chemists often react different amounts of material together to see what is formed. In an experiment a chemist reacted 1.198 g of carbon with 0.302 g of hydrogen. This made 1.500 g of our new material. What is the empirical formula?

Step 1:

Find the mole of each by taking the mass and dividing by the molar mass:

$$\frac{1.198\ \text{g}\ \cancel{C}}{12.01\ \text{g}\ \cancel{C}/\text{mol C}} = 0.0998\ \text{mol C} \qquad \frac{0.302\ \text{g}\ \cancel{H}}{1.01\ \text{g}\ \cancel{H}/\text{mol H}} = 0.299\ \text{mol H}$$

Finding the empirical formula-2

Step 2:

Take the larger number of moles and divide it by the smaller number of moles:

$$\frac{0.299\ \cancel{mol}\ H}{0.0998\ \cancel{mol}\ C} = \frac{3H}{1C}$$

Step 3:

Now take the coefficient from each element and use it as the subscript:

$$CH_3$$

From Empirical to Molecular Formula

- If we know the composition of a compound in terms of the masses (or mass percentages) of the elements present, we can calculate the empirical formula but not the molecular formula.
- To obtain the molecular formula we must know the molar mass.

Calculation of Molecular Formulas

- The molecular formula is always an integer multiple of the empirical formula.
  - For example, the sugar glucose has the empirical formula CH_2O and the molecular formula $C_6H_{12}O_6$.
- In general, we can represent the molecular formula in terms of the empirical formula as follows:

$$(\text{Empirical formula})_n = \text{molecular formula}$$

LA Pracitce

- $C_4H_5N_2O$

- MM is 194.14 g/mol

Think, Pair, Share Activity – Unit 8.9

- Turn to your neighbor and discuss why knowing the molecular formula is generally more useful than knowing the empirical formula.